

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$2 < x < 6$	B1	1.1b
		(1)	
(b)	States either $k > 8$ or $k < 0$	M1	3.1a
	States e.g. $\{k : k > 8\} \cup \{k : k < 0\}$	A1	2.5
		(2)	
(c)	Please see notes for alternatives		
	States $y = ax(x-6)^2$ or $f(x) = ax(x-6)^2$	M1	1.1b
	Substitutes (2,8) into $y = ax(x-6)^2$ and attempts to find a	dM1	3.1a
	$y = \frac{1}{4}x(x-6)^2$ or $f(x) = \frac{1}{4}x(x-6)^2$ o.e	A1	2.1
		(3)	
(6 marks)			
Notes: Watch for answers written by the question. If they are beside the question and in the answer space, the one in the answer space takes precedence			

(a)

B1: Deduces $2 < x < 6$ o.e. such as $x > 2, x < 6$ $x > 2$ and $x < 6$ $\{x : x > 2\} \cap \{x : x < 6\}$ $x \in (2, 6)$

Condone attempts in which set notation is incorrectly attempted but correct values can be seen

or implied E.g. $\{x > 2\} \cap \{x < 6\}$ $\{x > 2, x < 6\}$. Allow just the open interval $(2, 6)$

Do not allow for incorrect inequalities such as e.g. $x > 2$ or $x < 6$, $\{x : x > 2\} \cup \{x : x < 6\}$ $x \in [2, 6]$

(b)

M1: Establishes a correct method by finding one of the (correct) inequalities

States either $k > 8$ (condone $k \geq 8$) or $k < 0$ (condone $k \leq 0$)

Condone for this mark $y \leftrightarrow k$ or $f(x) \leftrightarrow k$ and $8 < k < 0$

A1: Fully correct solution in the form $\{k : k > 8\} \cup \{k : k < 0\}$ or $\{k | k > 8\} \cup \{k | k < 0\}$ either way around

but condone $\{k < 0\} \cup \{k > 8\}$, $\{k : k < 0 \cup k > 8\}$, $\{k < 0 \cup k > 8\}$. It is not necessary to

mention $\mathbb{R}$, e.g. $\{k : k \in \mathbb{R}, k > 8\} \cup \{k : k \in \mathbb{R}, k < 0\}$ Look for $\{ \}$ and $\cup$

Do not allow solutions not in set notation such as $k < 0$ or $k > 8$.

(c)

M1: Realises that the equation of C is of the form $y = ax(x-6)^2$. Condone with $a = 1$ for this mark.

So award for sight of $ax(x-6)^2$ even with $a = 1$

dM1: Substitutes (2,8) into the form $y = ax(x-6)^2$ and attempts to find the value for a .

It is dependent upon having an equation, which the ($y = \dots$) may be implied, of the correct form.

A1: Uses all of the information to form a correct equation for C $y = \frac{1}{4}x(x-6)^2$ o.e.

ISW after a correct answer. Condone $f(x) = \frac{1}{4}x(x-6)^2$ but not $C = \frac{1}{4}x(x-6)^2$.

Allow this to be written down for all 3 marks

Examples of alternative methods

Alternative I part (c):

Using the form $y = ax^3 + bx^2 + cx$ and setting up then solving simultaneous equations.

There are various versions of this but can be marked similarly

M1: Realises that the equation of C is of the form $y = ax^3 + bx^2 + cx$ and forms two equations in a , b and c . Condone with $a = 1$ for this mark.

Note that the form $y = ax^3 + bx^2 + cx + d$ is M0 until d is set equal to 0.

There are four equations that could be formed, only two are necessary for this mark.

Condone slips

Using $(6, 0) \Rightarrow 216a + 36b + 6c = 0$

Using $(2, 8) \Rightarrow 8a + 4b + 2c = 8$

Using $\frac{dy}{dx} = 0$ at $x = 2 \Rightarrow 12a + 4b + c = 0$

Using $\frac{dy}{dx} = 0$ at $x = 6 \Rightarrow 108a + 12b + c = 0$

dM1: Forms and solves three different equations, one of which must be using $(2, 8)$ to find values for a , b and c . A calculator can be used to solve the equations

A1: Uses all of the information to form a correct equation for C $y = \frac{1}{4}x^3 - 3x^2 + 9x$ o.e.

ISW after a correct answer. Condone $f(x) = \frac{1}{4}x^3 - 3x^2 + 9x$

Alternative II part (c)

Using the gradient and integrating

M1: Realises that the gradient of C is zero at 2 and 6 so sets $f'(x) = k(x-2)(x-6)$ oe and attempts to integrate. Condone with $k = 1$

dM1: Substitutes $x = 2, y = 8$ into $f(x) = k(\dots x^3 + \dots x + \dots)$ and finds a value for k

A1: Uses all of the information to form a correct equation for C $y = \frac{3}{4}\left(\frac{1}{3}x^3 - 4x^2 + 12x\right)$ o.e.

ISW after a correct answer. Condone $f(x) = \frac{3}{4}\left(\frac{1}{3}x^3 - 4x^2 + 12x\right)$

Question	Scheme	Marks	AOs
	Use of $\frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta}$	B1	2.1
	Uses the compound angle identity for $\sin(A+B)$ with $A=\theta$, $B=h$ $\Rightarrow \sin(\theta+h) = \sin\theta \cos h + \cos\theta \sin h$	M1	1.1b
	Achieves $\frac{\sin(\theta+h)-\sin\theta}{h} = \frac{\sin\theta \cos h + \cos\theta \sin h - \sin\theta}{h}$	A1	1.1b
	$= \frac{\sin h}{h} \cos\theta + \left(\frac{\cos h - 1}{h}\right) \sin\theta$	M1	2.1
	Uses $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$ Hence the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta} = \cos\theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos\theta$ *	A1*	2.5
(5 marks)			

Notes:

B1: States or implies that the gradient of the chord is $\frac{\sin(\theta+h)-\sin\theta}{h}$ or similar such as $\frac{\sin(\theta+\delta\theta)-\sin\theta}{\theta+\delta\theta-\theta}$ for a small h or $\delta\theta$

M1: Uses the compound angle identity for $\sin(A+B)$ with $A=\theta$, $B=h$ or $\delta\theta$

A1: Obtains $\frac{\sin\theta \cos h + \cos\theta \sin h - \sin\theta}{h}$ or equivalent

M1: Writes their expression in terms of $\frac{\sin h}{h}$ and $\frac{\cos h - 1}{h}$

A1*: Uses correct language to explain that $\frac{dy}{d\theta} = \cos\theta$

For this method they should use all of the given statements $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$,

$\frac{\cos h - 1}{h} \rightarrow 0$ meaning that the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta} = \cos\theta$

and therefore the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos\theta$

Question	Scheme	Marks	AOs
alt	Use of $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta}$	B1	2.1
	Sets $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \frac{\sin\left(\theta + \frac{h}{2} + \frac{h}{2}\right) - \sin\left(\theta + \frac{h}{2} - \frac{h}{2}\right)}{h}$ and uses the compound angle identity for $\sin(A+B)$ and $\sin(A-B)$ with $A = \theta + \frac{h}{2}$, $B = \frac{h}{2}$	M1	1.1b
	Achieves $\frac{\sin(\theta+h) - \sin \theta}{h} = \frac{\left[\sin\left(\theta + \frac{h}{2}\right)\cos\left(\frac{h}{2}\right) + \cos\left(\theta + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right] - \left[\sin\left(\theta + \frac{h}{2}\right)\cos\left(\frac{h}{2}\right) - \cos\left(\theta + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right]}{h}$	A1	1.1b
	$= \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \times \cos\left(\theta + \frac{h}{2}\right)$	M1	2.1
	Uses $h \rightarrow 0, \frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ and $\cos\left(\theta + \frac{h}{2}\right) \rightarrow \cos \theta$ Therefore the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$ *	A1*	2.5
(5 marks)			

Additional notes:

A1*: Uses correct language to explain that $\frac{dy}{d\theta} = \cos \theta$. For this method they should use the

(adapted) given statement $h \rightarrow 0, \frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ with $\cos\left(\theta + \frac{h}{2}\right) \rightarrow \cos \theta$

meaning that the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and therefore the gradient of the

chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{dy}{dx} = \frac{(x+1)^2 \times (10x+10) - (5x^2+10x) \times 2(x+1)}{(x+1)^4}$	M1	This mark is given for an attempt to differentiate the expression for y
		A1	This mark is given for correctly differentiating the expression for y
	$\frac{dy}{dx} = \frac{(x+1) \times (10x+10) - (5x^2+10x) \times 2}{(x+1)^3}$	M1	This mark is given for cancelling the expression through by $(x+1)$
	$\frac{dy}{dx} = \frac{10}{(x+1)^3}$	A1	This mark is given for a fully correct expression for $\frac{dy}{dx}$
(b)	If $A > 0$ and $n = 1, 3$ then $x < -1$	B1	This mark is given for deducing that $\frac{dy}{dx} < 0 \Rightarrow x < -1$.

Q4.

Question	Scheme	Marks	AOs
(a)(i)	$\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$	M1 A1	1.1b 1.1b
(ii)	$\frac{d^2y}{dx^2} = 60x^2 - 144x + 84$	A1ft	1.1b
		(3)	

(b)(i)	$x = 1 \Rightarrow \frac{dy}{dx} = 20 - 72 + 84 - 32$	M1	1.1b
	$\frac{dy}{dx} = 0$ so there is a stationary point at $x = 1$	A1	2.1
Alternative for (b)(i)			
	$20x^3 - 72x^2 + 84x - 32 = 4(x-1)^2(5x-8) = 0 \Rightarrow x = \dots$	M1	1.1b
	When $x = 1$, $\frac{dy}{dx} = 0$ so there is a stationary point	A1	2.1
(b)(ii)	Note that in (b)(ii) there are no marks for <u>just</u> evaluating $\left(\frac{d^2y}{dx^2}\right)_{x=1}$		
	E.g. $\left(\frac{d^2y}{dx^2}\right)_{x=0.8} = \dots \quad \left(\frac{d^2y}{dx^2}\right)_{x=1.2} = \dots$	M1	2.1
	$\left(\frac{d^2y}{dx^2}\right)_{x=0.8} > 0, \quad \left(\frac{d^2y}{dx^2}\right)_{x=1.2} < 0$ Hence point of inflection	A1	2.2a
		(4)	
Alternative 1 for (b)(ii)			
	$\left(\frac{d^2y}{dx^2}\right)_{x=1} = 60x^2 - 144x + 84 = 0$ (is inconclusive) $\left(\frac{d^3y}{dx^3}\right) = 120x - 144 \Rightarrow \left(\frac{d^3y}{dx^3}\right)_{x=1} = \dots$	M1	2.1
	$\left(\frac{d^2y}{dx^2}\right)_{x=1} = 0$ and $\left(\frac{d^3y}{dx^3}\right)_{x=1} \neq 0$ Hence point of inflection	A1	2.2a
Alternative 2 for (b)(ii)			
	E.g. $\left(\frac{dy}{dx}\right)_{x=0.8} = \dots \quad \left(\frac{dy}{dx}\right)_{x=1.2} = \dots$	M1	2.1
	$\left(\frac{dy}{dx}\right)_{x=0.8} < 0, \quad \left(\frac{dy}{dx}\right)_{x=1.2} < 0$ Hence point of inflection	A1	2.2a
(7 marks)			

(a)(i)

M1: $x^n \rightarrow x^{n-1}$ for at least one power of x

A1: $\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$

(a)(ii)

A1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$

(b)(i)

M1: Substitutes $x = 1$ into their $\frac{dy}{dx}$

A1: Obtains $\frac{dy}{dx} = 0$ following a correct derivative and makes a conclusion which can be minimal

e.g. tick, QED etc. which may be in a preamble e.g. stationary point when $\frac{dy}{dx} = 0$ and then

shows $\frac{dy}{dx} = 0$

Alternative:

M1: Attempts to solve $\frac{dy}{dx} = 0$ by factorisation. This may be by using the factor of $(x - 1)$ or possibly using a calculator to find the roots and showing the factorisation. Note that they may divide by 4 before factorising which is acceptable. Need to either see either $4(x-1)^2(5x-8)$ or $(x-1)^2(5x-8)$ for the factorisation or $x = \frac{8}{5}$ and $x = 1$ seen as the roots.

A1: Obtains $x = 1$ and makes a conclusion as above

(b)(ii)

M1: Considers the value of the second derivative either side of $x = 1$. Do not be too concerned with the interval for the method mark.

(NB $\frac{d^2y}{dx^2} = (x-1)(60x-84)$ so may use this factorised form when considering $x < 1$, $x > 1$ for sign change of second derivative)

A1: Fully correct work including a correct $\frac{d^2y}{dx^2}$ with a reasoned conclusion indicating that the stationary point is a point of inflection. Sufficient reason is e.g. "sign change"/" > 0 , < 0 ". If values are given they should be correct (but be generous with accuracy) but also just allow " > 0 " and " < 0 " provided they are correctly paired. The interval must be where $x < 1.4$

Alternative 1 for (b)(ii)

M1: Shows that second derivative at $x = 1$ is zero and then finds the third derivative at $x = 1$

A1: Fully correct work including a correct $\frac{d^2y}{dx^2}$ with a reasoned conclusion indicating that stationary point is a point of inflection. Sufficient reason is " $\neq 0$ " but must follow a correct third derivative and a correct value if evaluated. For reference $\left(\frac{d^3y}{dx^3}\right)_{x=1} = -24$

Alternative 2 for (b)(ii)

M1: Considers the value of the first derivative either side of $x = 1$. Do not be too concerned with the interval for the method mark.

A1: Fully correct work with a reasoned conclusion indicating that stationary point is a point of inflection. Sufficient reason is e.g. "same sign"/"both negative"/" < 0 , < 0 ". If values are given they should be correct (but be generous with accuracy). The interval must be where $x < 1.4$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$f'(x)$	-32	-24.3	-17.92	-12.74	-8.64	-5.5	-3.2	-1.62	-0.64	-0.14	0
$f''(x)$	84	70.2	57.6	46.2	36	27	19.2	12.6	7.2	3	0

x	1.1	1.2	1.3	1.4	1.5	1.6	1.7
$f'(x)$	-0.1	-0.32	-0.54	-0.64	-0.5	0	0.98
$f''(x)$	-1.8	-2.4	-1.8	0	3	7.2	12.6

Q5.

Question	Scheme	Marks	AOs
(a)	$y = x^3 - 10x^2 + 27x - 23 \Rightarrow \frac{dy}{dx} = 3x^2 - 20x + 27$	B1	1.1b
	$\left(\frac{dy}{dx}\right)_{x=5} = 3 \times 5^2 - 20 \times 5 + 27 (= 2)$	M1	1.1b
	$y + 13 = 2(x - 5)$	M1	2.1
	$y = 2x - 23$	A1	1.1b
		(4)	
(b)	Both C and l pass through $(0, -23)$ and so C meets l again on the y -axis	B1	2.2a
		(1)	



(c)	$\pm \int (x^3 - 10x^2 + 27x - 23 - (2x - 23)) \, dx$ $= \pm \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right)$	M1 A1ft	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right]_0^5$ $= \left(\frac{625}{4} - \frac{1250}{3} + \frac{625}{2} \right) (-0)$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b
		(4)	
(c) Alternative 1:			
	$\pm \int (x^3 - 10x^2 + 27x - 23) \, dx$ $= \pm \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right)$	M1 A1	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right]_0^5 + \frac{1}{2} \times 5(23+13)$ $= -\frac{455}{12} + 90$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b
(c) Alternative 2:			
	$\int (x^3 - 10x^2 + 27x) \, dx = \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right)$	M1 A1	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right]_0^5 - \frac{1}{2} \times 5 \times 10$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b

(9 marks)

Notes

(a)

B1: Correct derivative

M1: Substitutes $x = 5$ into their derivative. This may be implied by their value for $\frac{dy}{dx}$ M1: Fully correct straight line method using $(5, -13)$ and their $\frac{dy}{dx}$ at $x = 5$

A1: cao. Must see the full equation in the required form.

(b)

B1: Makes a suitable deduction.

Alternative via equating l and C and factorising e.g.

$$x^3 - 10x^2 + 27x - 23 = 2x - 23$$

$$x^3 - 10x^2 + 25x = 0$$

$$x(x^2 - 10x + 25) = 0 \Rightarrow x = 0$$

So they meet on the y -axis



(c)

M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $\pm C - l$

A1ft: Correct integration in any form which may be simplified or unsimplified. (follow through their equation from (a))

If they attempt as 2 separate integrals e.g. $\int (x^3 - 10x^2 + 27x - 23) dx - \int (2x - 23) dx$ then

award this mark for the correct integration of the curve as in the alternative.

If they combine the curve with the line first then the subsequent integration must be correct or a correct ft for their line and allow for $\pm C - l$

dm1: Fully correct strategy for the area. Award for use of 5 as the limit and condone the omission of the $- 0$. **Depends on the first method mark.**

A1: Correct exact value

Alternative 1:M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $\pm C$ A1: Correct integration for $\pm C$

dm1: Fully correct strategy for the area e.g. correctly attempts the area of the trapezium and subtracts the area enclosed between the curve and the x -axis. Need to see the use of 5 as the limit condoning the omission of the $- 0$ and a correct attempt at the trapezium and the subtraction.

May see the trapezium area attempted as $\int (2x - 23) dx$ in which case the integration and

use of the limits needs to be correct or correct follow through for their straight line equation.

Depends on the first method mark.

A1: Correct exact value

Note if they do $l - C$ rather than $C - l$ and the working is otherwise correct allow full marks if their final answer is given as a positive value. E.g. correct work with $l - C$ leading to $-\frac{625}{12}$ and

then e.g. hence area is $\frac{625}{12}$ is acceptable for full marks.

If the answer is left as $-\frac{625}{12}$ then score A0

Alternative 2:M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $(C + 23)$ A1: Correct integration for $(C + 23)$

dm1: Fully correct strategy for the area e.g. correctly attempts the area of the triangle and subtracts from the area under the curve

Need to see the use of 5 as the limit condoning the omission of the $- 0$ and a correct attempt at the triangle and the subtraction.

Depends on the first method mark.

Q6.

Question	Scheme	Marks	AOs
	Attempts the product and chain rule on $y = x(2x+1)^4$	M1	2.1
	$\frac{dy}{dx} = (2x+1)^4 + 8x(2x+1)^3$	A1	1.1b
	Takes out a common factor $\frac{dy}{dx} = (2x+1)^3 \{(2x+1) + 8x\}$	M1	1.1b
	$\frac{dy}{dx} = (2x+1)^3(10x+1) \Rightarrow n=3, A=10, B=1$	A1	1.1b
(4 marks)			
Notes:			
M1: Applies the product rule to reach $\frac{dy}{dx} = (2x+1)^4 + Bx(2x+1)^3$ A1: $\frac{dy}{dx} = (2x+1)^4 + 8x(2x+1)^3$ M1: Takes out a common factor of $(2x+1)^3$ A1: The form of this answer is given. Look for $\frac{dy}{dx} = (2x+1)^3(10x+1) \Rightarrow n=3, A=10, B=1$			

Q7.

Question	Scheme	Marks	AOs
(a)	(i) $\frac{dy}{dx} = 12x^3 - 24x^2$	M1	1.1b
		A1	1.1b
	(ii) $\frac{d^2y}{dx^2} = 36x^2 - 48x$	A1ft	1.1b
		(3)	
(b)	Substitutes $x = 2$ into their $\frac{dy}{dx} = 12 \times 2^3 - 24 \times 2^2$	M1	1.1b
	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point"	A1	2.1
		(2)	
(c)	Substitutes $x = 2$ into their $\frac{d^2y}{dx^2} = 36 \times 2^2 - 48 \times 2$	M1	1.1b
	$\frac{d^2y}{dx^2} = 48 > 0$ and states "hence the stationary point is a minimum"	A1ft	2.2a
		(2)	
(7 marks)			

Notes:	
(a)(i)	
M1:	Differentiates to a cubic form
A1:	$\frac{dy}{dx} = 12x^3 - 24x^2$
(a)(ii)	
A1ft:	Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx} = 36x^2 - 48x$
(b)	
M1:	Substitutes $x = 2$ into their $\frac{dy}{dx}$
A1:	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" All aspects of the proof must be correct
(c)	
M1:	Substitutes $x = 2$ into their $\frac{d^2y}{dx^2}$
	Alternatively calculates the gradient of C either side of $x = 2$
A1ft:	For a correct calculation, a valid reason and a correct conclusion.
	Follow through on an incorrect $\frac{d^2y}{dx^2}$

Q8.

Question	Scheme	Marks	AOs
(a)	Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$	M1	1.1b
	$\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} \quad (= 2\sqrt{3} \cos t)$	A1	1.1b
	(2)		
(b)	Substitutes $t = \frac{2\pi}{3}$ in $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} = (-\sqrt{3})$	M1	2.1
	Uses gradient of normal $= -\frac{1}{dy/dx} = \left(\frac{1}{\sqrt{3}}\right)$	M1	2.1
	Coordinates of $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$	B1	1.1b
	Correct form of normal $y + \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x+1)$	M1	2.1
	Completes proof $\Rightarrow 2x - 2\sqrt{3}y - 1 = 0$ *	A1*	1.1b
	(5)		
(c)	Substitutes $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$	M1	3.1a
	Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic in $\cos t$	M1	3.1a
	$\Rightarrow 12\cos^2 t - 4\cos t - 5 = 0$	A1	1.1b
	Finds $\cos t = \frac{5}{6}, \frac{1}{2}$	M1	2.4
	Substitutes their $\cos t = \frac{5}{6}$ into $x = 2\cos t, y = \sqrt{3} \cos 2t,$	M1	1.1b
	$Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$	A1	1.1b
	(6)		
(13 marks)			

**Notes:**

(a)

M1: Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ and achieves a form $k \frac{\sin 2t}{\sin t}$ Alternatively candidates may apply the double angle identity for $\cos 2t$ and achieve a form $k \frac{\sin t \cos t}{\sin t}$

A1: Scored for a correct answer, either $\frac{\sqrt{3} \sin 2t}{\sin t}$ or $2\sqrt{3} \cos t$

(b)

M1: For substituting $t = \frac{2\pi}{3}$ in their $\frac{dy}{dx}$ which must be in terms of t

M1: Uses the gradient of the normal is the negative reciprocal of the value of $\frac{dy}{dx}$. This may be seen in the equation of l .

B1: States or uses (in their tangent or normal) that $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$

M1: Uses their numerical value of $-1/\frac{dy}{dx}$ with their $\left(-1, -\frac{\sqrt{3}}{2}\right)$ to form an equation of the normal at P

A1*: This is a proof and all aspects need to be correct. Correct answer only $2x - 2\sqrt{3}y - 1 = 0$

(c)

M1: For substituting $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$ to produce an equation in t . Alternatively candidates could use $\cos 2t = 2\cos^2 t - 1$ to set up an equation of the form $y = Ax^2 + B$.

M1: Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic equation in $\cos t$
In the alternative method it is for combining their $y = Ax^2 + B$ with $2x - 2\sqrt{3}y - 1 = 0$ to get an equation in just one variable

A1: For the correct quadratic equation $12\cos^2 t - 4\cos t - 5 = 0$
Alternatively the equations in x and y are $3x^2 - 2x - 5 = 0$ $12\sqrt{3}y^2 + 4y - 7\sqrt{3} = 0$

M1: Solves the quadratic equation in $\cos t$ (or x or y) and rejects the value corresponding to P .

M1: Substitutes their $\cos t = \frac{5}{6}$ or their $t = \arccos\left(\frac{5}{6}\right)$ in $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$

If a value of x or y has been found it is for finding the other coordinate.

A1: $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$. Allow $x = \frac{5}{3}, y = \frac{7}{18}\sqrt{3}$ but do not allow decimal equivalents.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{dx}{dy} = 8 \cos 2y \Rightarrow \frac{dy}{dx} = \frac{1}{8 \cos 2y}$	M1	This mark is given for differentiating and inverting
	At (0, 0), $\frac{dy}{dx} = \frac{1}{8}$	A1	This mark is given for finding $\frac{dy}{dx}$ when $y = 0$
(b)(i)	$\sin 2y \approx 2y \Rightarrow x \approx 8y$	B1	This mark is given for finding an approximation for x
(b)(ii)	When x and y are small, $x = 4 \sin 2y$ approximates to the line $x = 8y$	B1	This mark is given for a valid explanation of the relationship between x and y when both are small
(c)	$\sin^2 2y + \cos^2 2y = 1$ $\Rightarrow \cos^2 2y = 1 - \sin^2 2y$ $x = 4 \sin 2y \Rightarrow \sin^2 2y = \left(\frac{x}{4}\right)^2$	M1	This mark is given for a method to use find an expression for $\sin^2 2y$ in terms of x
	$\frac{dy}{dx} = \frac{1}{8 \cos 2y} = \frac{1}{8 \sqrt{1 - \left(\frac{x}{4}\right)^2}}$	A1	This mark is given for an unsimplified expression for $\frac{dy}{dx}$
	$\frac{dy}{dx} = \frac{1}{2\sqrt{16 - x^2}}$	A1	This mark is given for a fully correct answer with $a = 2$ and $b = 16$

Q10.

Question	Scheme	Marks	AOs
	$N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \quad \frac{dN}{dt} = \frac{N(300 - N)}{1200}$		
(a)	90	B1	3.4
		(1)	
(b) Way 1	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\Rightarrow \frac{dN}{dt} = \frac{900(0.25) \left(\left(\frac{900}{N} - 3 \right) \right)}{\left(\frac{900}{N} \right)^2}$	dM1	2.1
	correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} *$	A1*	1.1b
		(4)	
(b) Way 2	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\frac{N(300 - N)}{1200} = \frac{\left(\frac{900}{3 + 7e^{-0.25t}} \right) \left(300 - \frac{900}{3 + 7e^{-0.25t}} \right)}{1200}$	dM1	2.1
	$\text{LHS} = \frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \text{ o.e.,}$ $\text{RHS} = \frac{900(300(3 + 7e^{-0.25t}) - 900)}{1200(3 + 7e^{-0.25t})^2} = \frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \text{ o.e.}$ and states hence $\frac{dN}{dt} = \frac{N(300 - N)}{1200}$ (or LHS = RHS) *	A1*	1.1b
		(4)	
(c)	Deduces $N = 150$ (can be implied)	B1	2.2a
	so $150 = \frac{900}{3 + 7e^{-0.25T}} \Rightarrow e^{-0.25T} = \frac{3}{7}$	M1	3.4
	$T = -4 \ln\left(\frac{3}{7}\right)$ or $T = \text{awrt } 3.4$ (months)	dM1	1.1b
		A1	1.1b
		(4)	
(d)	either one of 299 or 300	B1	3.4
		(1)	
(10 marks)			

Notes for Question	
(b)	
M1:	Attempts to differentiate using - the chain rule to give $\frac{dN}{dt} = \pm Ae^{-0.25t}(3+7e^{-0.25t})^{-2}$ or $\frac{\pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}$ o.e. - the quotient rule to give $\frac{dN}{dt} = \frac{(3+7e^{-0.25t})(0) \pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}$ - implicit differentiation to give $N(3+7e^{-0.25t}) = 900 \Rightarrow (3+7e^{-0.25t})\frac{dN}{dt} \pm ANe^{-0.25t} = 0$, o.e. where $A \neq 0$
Note:	Condone a slip in copying $(3+7e^{-0.25t})$ for the M mark
A1:	A correct differentiation statement
Note:	Implicit differentiation gives $(3+7e^{-0.25t})\frac{dN}{dt} - 1.75Ne^{-0.25t} = 0$
dM1:	Way 1: Complete attempt, by eliminating t , to form an equation linking $\frac{dN}{dt}$ and N only Way 2: Complete substitution of $N = \frac{900}{3+7e^{-0.25t}}$ into $\frac{dN}{dt} = \frac{N(300-N)}{1200}$
Note:	Way 1: e.g. substitutes $3+7e^{-0.25t} = \frac{900}{N}$ and $e^{-0.25t} = \frac{900}{N} - 3$ or substitutes $e^{-0.25t} = \frac{\frac{900}{N} - 3}{7}$ into their $\frac{dN}{dt} = \dots$ to form an equation linking $\frac{dN}{dt}$ and N
A1*:	Way 1: Correct algebra leading to $\frac{dN}{dt} = \frac{N(300-N)}{1200}$ * Way 2: See scheme
(c)	
B1:	Deduces or shows that $\frac{dN}{dt}$ is maximised when $N=150$
M1:	Uses the model $N = \frac{900}{3+7e^{-0.25t}}$ with their $N=150$ and proceeds as far as $e^{-0.25t} = k, k > 0$ or $e^{0.25t} = k, k > 0$. Condone $t \equiv T$
dM1:	Correct method of using logarithms to find a value for T . Condone $t \equiv T$
A1:	see scheme
Note:	$\frac{d^2N}{dt^2} = \frac{dN}{dt} \left(\frac{300}{1200} - \frac{2N}{1200} \right) = 0 \Rightarrow N=150$ is acceptable for B1
Note:	Ignore units for T
Note:	Applying $300 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ or $0 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ is M0 dM0 A0
Note:	M1 dM1 can only be gained in (c) by using an N value in the range $90 < N < 300$
(d)	
B1:	300 (or accept 299)



Question	Scheme	Marks	AOs
	$N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \frac{dN}{dt} = \frac{N(300 - N)}{1200}$		
(b) Way 3	$\int \frac{1}{N(300 - N)} dN = \int \frac{1}{1200} dt$	M1	2.1
	$\int \frac{1}{300} \left(\frac{1}{N} + \frac{1}{300 - N} \right) dN = \int \frac{1}{1200} dt$	A1	1.1b
	$\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$		
	$\{t = 0, N = 90 \Rightarrow\} c = \frac{1}{300} \ln(90) - \frac{1}{300} \ln(210) \Rightarrow c = \frac{1}{300} \ln\left(\frac{3}{7}\right)$ $\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t + \frac{1}{300} \ln\left(\frac{3}{7}\right)$ $\ln N - \ln(300 - N) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right)$ $\ln\left(\frac{N}{300 - N}\right) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right) \Rightarrow \frac{N}{300 - N} = \frac{3}{7} e^{\frac{1}{4}t}$	dM1	2.1
	$7N = 3e^{\frac{1}{4}t}(300 - N) \Rightarrow 7N + 3Ne^{\frac{1}{4}t} = 900e^{\frac{1}{4}t}$ $N(7 + 3e^{\frac{1}{4}t}) = 900e^{\frac{1}{4}t} \Rightarrow N = \frac{900e^{\frac{1}{4}t}}{7 + 3e^{\frac{1}{4}t}} \Rightarrow N = \frac{900}{3 + 7e^{-0.25t}} *$	A1*	1.1b
	(4)		
(b) Way 4	$N(3 + 7e^{-0.25t}) = 900 \Rightarrow e^{-0.25t} = \frac{1}{7} \left(\frac{900}{N} - 3 \right) \Rightarrow e^{-0.25t} = \frac{900 - 3N}{7N}$ $\Rightarrow t = -4(\ln(900 - 3N) - \ln(7N))$ $\Rightarrow \frac{dt}{dN} = -4 \left(\frac{-3}{900 - 3N} - \frac{7}{7N} \right)$	M1	2.1
	$\frac{dt}{dN} = 4 \left(\frac{1}{300 - N} + \frac{1}{N} \right) \Rightarrow \frac{dt}{dN} = 4 \left(\frac{N + 300 - N}{N(300 - N)} \right)$	A1	1.1b
	$\frac{dt}{dN} = \left(\frac{1200}{N(300 - N)} \right) \Rightarrow \frac{dN}{dt} = \frac{N(300 - N)}{1200} *$	dM1	2.1
		A1*	1.1b
	(4)		

Notes for Question Continued

(b) Way 3	
M1:	Separates the variables, an attempt to form and apply partial fractions and integrates to give $\ln$ terms = $kt \{+c\}$, $k \neq 0$, with or without a constant of integration c
A1:	$\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$ or equivalent with or without a constant of integration c
dM1:	Uses $t = 0, N = 90$ to find their constant of integration and obtains an expression of the form $\lambda e^{\frac{1}{4}t} = f(N)$; $\lambda \neq 0$ or $\lambda e^{-\frac{1}{4}t} = f(N)$; $\lambda \neq 0$
A1*:	Correct manipulation leading to $N = \frac{900}{3 + 7e^{-0.25t}} *$
(b) Way 4	
M1:	Valid attempt to make t the subject, followed by an attempt to find two $\ln$ derivatives, condoning sign errors and constant errors.
A1:	$\frac{dt}{dN} = -4 \left(\frac{-3}{900 - 3N} - \frac{7}{7N} \right)$ or equivalent
dM1:	Forms a common denominator to combine their fractions
A1*:	Correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} *$

Q11.

Question	Scheme	Marks	AOs
	$C: y = x \ln x$; l is a normal to C at $P(e, e)$ Let x_A be the x -coordinate of where l cuts the x -axis		
	$\frac{dy}{dx} = \ln x + x \left(\frac{1}{x} \right) \quad \{= 1 + \ln x\}$	M1	2.1
		A1	1.1b
	$x = e, m_T = 2 \Rightarrow m_N = -\frac{1}{2} \Rightarrow y - e = -\frac{1}{2}(x - e)$ $y = 0 \Rightarrow -e = -\frac{1}{2}(x - e) \Rightarrow x = \dots$	M1	3.1a
	l meets x -axis at $x = 3e$ (allow $x = 2e + e \ln e$)	A1	1.1b
	{Areas:} either $\int_1^e x \ln x \, dx = [\dots]_1^e = \dots$ or $\frac{1}{2}((\text{their } x_A) - e)e$	M1	2.1
	$\left\{ \int x \ln x \, dx = \right\} \frac{1}{2} x^2 \ln x - \int \frac{1}{x} \cdot \left(\frac{x^2}{2} \right) \{dx\}$	M1	2.1
	$\left\{ = \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x \{dx\} \right\} = \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2$	dM1	1.1b
		A1	1.1b
	$\text{Area}(R_1) = \int_1^e x \ln x \, dx = [\dots]_1^e = \dots$; $\text{Area}(R_2) = \frac{1}{2}((\text{their } x_A) - e)e$ and so, $\text{Area}(R) = \text{Area}(R_1) + \text{Area}(R_2) \quad \{= \frac{1}{4}e^2 + \frac{1}{4} + e^2\}$	M1	3.1a
	$\text{Area}(R) = \frac{5}{4}e^2 + \frac{1}{4}$	A1	1.1b
		(10)	

Notes for Question	
M1:	Differentiates by using the product rule to give $\ln x + x(\text{their } g'(x))$, where $g(x) = \ln x$
A1:	Correct differentiation of $y = x \ln x$, which can be un-simplified or simplified
M1:	Complete strategy to find the x coordinate where their normal to C at $P(e, e)$ meets the x -axis i.e. Sets $y=0$ in $y-e = m_N(x-e)$ to find $x = \dots$
Note:	m_T is found by using calculus and $m_N \neq m_T$
A1:	l meets x -axis at $x = 3e$, allowing un-simplified values for x such as $x = 2e + e \ln e$
Note:	Allow $x = \text{awrt } 8.15$
M1:	Scored for either - Area under curve $= \int_1^e x \ln x \, dx = [\dots]_1^e = \dots$, with limits of e and 1 and some attempt to substitute these and subtract - or Area under line $= \frac{1}{2}((\text{their } x_A) - e)e$, with a valid attempt to find x_A
M1:	Integration by parts the correct way around to give $Ax^2 \ln x - \int B \left(\frac{x^2}{x} \right) \{dx\}$; $A \neq 0, B > 0$
dM1:	dependent on the previous M mark Integrates the second term to give $\pm \lambda x^2$; $\lambda \neq 0$
A1:	$\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2$
M1:	Complete strategy of finding the area of R by finding the sum of two key areas. See scheme.
A1:	$\frac{5}{4}e^2 + \frac{1}{4}$
Note:	Area(R_2) can also be found by integrating the line l between limits of e and their x_A i.e. $\text{Area}(R_2) = \int_e^{\text{their } x_A} \left(-\frac{1}{2}x + \frac{3}{2}e \right) dx = [\dots]_e^{\text{their } x_A} = \dots$
Note:	Calculator approach with no algebra, differentiation or integration seen: - Finding l cuts through the x-axis at awrt 8.15 is 2nd M1 2nd A1 - Finding area between curve and the x-axis between $x=1$ and $x=e$ to give awrt 2.10 is 3rd M1 - Using the above information (must be seen) to apply $\text{Area}(R) = 2.0972\dots + 7.3890\dots = 9.4862\dots$ is final M1 Therefore, a maximum of 4 marks out of the 10 available.

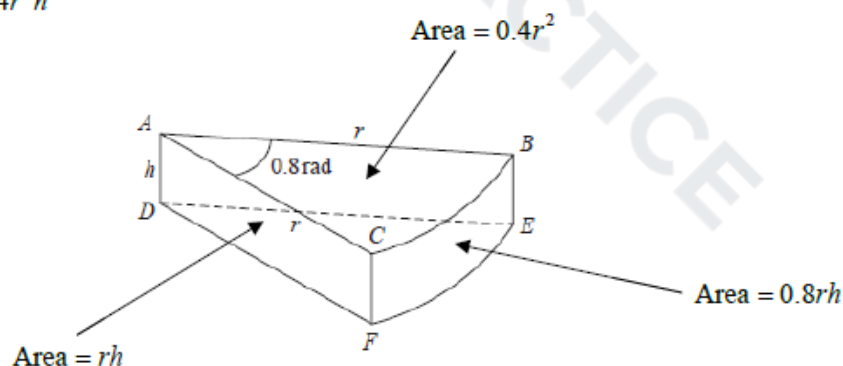
Q12.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$6 = \pi r^2 h + \frac{2}{3} \pi r^3$	B1	This mark is given for a method to find the volume of the cylinder and the semi-hemisphere
	$A = 3\pi r^2 + 2\pi \left(\frac{6 - \frac{2}{3}\pi r^3}{\pi r} \right)$	M1	This mark is given for a method to find the surface area of the tank
		A1	This mark is given for finding an expression for the surface area of the tank
	$A = 3\pi r^2 + \frac{12}{r} - \frac{4\pi r^2}{3} = \frac{12}{r} + \frac{5\pi r^2}{3}$	A1	This mark is given for a fully correct proof to show the surface area of the tank as required
(b)	$A = \frac{12}{r} + \frac{5\pi r^2}{3} \Rightarrow \frac{dA}{dr} = -\frac{12}{r^2} + \frac{10\pi r}{3}$	M1	This mark is given for a method to differentiate to find r
		A1	This mark is given for accurately differentiating to find r
	When $\frac{dA}{dr} = 0$, $-\frac{12}{r^2} + \frac{10\pi r}{3} = 0$ $r^3 = \frac{18}{5\pi}$	M1	This mark is given for a method to set $\frac{dA}{dr} = 0$ to find a value for r
	$r = 1.046$	A1	This mark is given for finding the radius for which the surface area is a minimum
(c)	$A = \frac{12}{1.046} + \frac{5\pi(1.046)^2}{3}$	M1	This mark is given for a method to substitute a value for r
	$A = 17 \text{ m}^2$	A1	This mark is given for correctly finding the minimum surface area of the tank (to the nearest integer)
(Total 10 marks)			

Q13.

Question	Scheme	Marks	AOs
(a)	Sets up an allowable equation using volume = 240 E.g. $\frac{1}{2}r^2 \times 0.8h = 240 \Rightarrow h = \frac{600}{r^2}$ o.e.	M1 A1	3.4 1.1b
	Attempts to substitute their $h = \frac{600}{r^2}$ into $(S =) \frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + 2rh + 0.8rh$	dM1	3.4
	$S = 0.8r^2 + 2.8rh = 0.8r^2 + 2.8 \times \frac{600}{r} = 0.8r^2 + \frac{1680}{r}$ *	A1*	2.1
	(4)		
(b)	$\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$	M1 A1	3.1a 1.1b
	Sets $\frac{dS}{dr} = 0 \Rightarrow r^3 = 1050$ $r = \text{awrt } 10.2$	dM1 A1	2.1 1.1b
	(4)		
(c)	Attempts to substitute their positive r into $\left(\frac{d^2S}{dr^2}\right) = 1.6 + \frac{3360}{r^3}$ and considers its value or sign	M1	1.1b
	E.g. Correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$ with $\frac{d^2S}{dr^2} \bigg _{r=10.2} = 5 > 0$ proving a minimum value of S	A1	1.1b
	(2)		
(10 marks)			
Notes:			

$$\text{Volume} = 0.4r^2h$$



$$\text{Total surface area} = 2rh + 0.8r^2 + 0.8rh$$

(a)

M1: Attempts to use the fact that the volume of the toy is 240 cm^3

Sight of $\frac{1}{2}r^2 \times 0.8 \times h = 240$ leading to $h = \dots$ or $rh = \dots$ scores this mark

But condone an equation of the correct form so allow for $kr^2h = 240 \Rightarrow h = \dots$ or $rh = \dots$

A1: A correct expression for $h = \frac{600}{r^2}$ or $rh = \frac{600}{r}$ which may be left unsimplified.

This may be implied when you see an expression for S or part of S E.g. $2rh = 2r \times \frac{600}{r^2}$

dM1: Attempts to substitute their $h = \frac{a}{r^2}$ o.e. such as $hr = \frac{a}{r}$ into a correct expression for S

Sight of $\frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + rh + rh + 0.8rh$ with an appropriate substitution

Simplified versions such as $0.8r^2 + 2rh + 0.8rh$ used with an appropriate substitution is fine.

A1*: Correct work leading to the given result.

$S =$, $SA =$ or surface area = must be seen at least once in the correct place

The method must be made clear so expect to see evidence. For example

$S = 0.8r^2 + 2rh + 0.8rh \Rightarrow S = 0.8r^2 + 2r \times \frac{600}{r^2} + 0.8r \times \frac{600}{r^2} \Rightarrow S = 0.8r^2 + \frac{1680}{r}$ would be fine.

(b) There is no requirement to see $\frac{dS}{dr}$ in part (b). It may even be called $\frac{dy}{dx}$.

M1: Achieves a derivative of the form $pr \pm \frac{q}{r^2}$ where p and q are non-zero constants

A1: Achieves $\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$

dM1: Sets or implies that their $\frac{dS}{dr} = 0$ and proceeds to $mr^3 = n$, $m \times n > 0$. It is dependent upon a

correct attempt at differentiation. This mark may be implied by a correct answer to their $pr - \frac{q}{r^2} = 0$

A1: $r = \text{awrt } 10.2$ or $\sqrt[3]{1050}$

(c)

M1: Attempts to substitute their positive r (found in (b)) into $\left(\frac{d^2S}{dr^2}\right) = e \pm \frac{f}{r^3}$ where e and f are non zero

and finds its value or sign.

Alternatively considers the sign of $\left(\frac{d^2S}{dr^2}\right) = e \pm \frac{f}{r^3}$ (at their positive r found in (b))

Condone the $\frac{d^2S}{dr^2}$ to be $\frac{d^2y}{dx^2}$ or being absent, but only for this mark.

A1: States that $\frac{d^2S}{dr^2}$ or $S'' = 1.6 + \frac{3360}{r^3} = \text{awrt } 5 > 0$ proving a minimum value of S

This is dependent upon having achieved $r = \text{awrt } 10$ and a correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$

It can be argued without finding the value of $\frac{d^2S}{dr^2}$. E.g. $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3} > 0$ as $r > 0$, so

minimum value of S . For consistency it is also dependent upon having achieved $r = \text{awrt } 10$

Do NOT allow $\frac{d^2y}{dx^2}$ for this mark

Q14.

Question	Scheme	Marks	AOs
	$\frac{d}{d\theta}(\cos \theta) = -\sin \theta$; as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$		
	$\frac{\cos(\theta + h) - \cos \theta}{h}$	B1	2.1
	$= \frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h}$	M1	1.1b
		A1	1.1b
	$= -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$		
	As $h \rightarrow 0$, $-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta \rightarrow -1 \sin \theta + 0 \cos \theta$	dM1	2.1
	so $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$ *	A1*	2.5
		(5)	
(5 marks)			

Notes for Question	
B1:	Gives the correct fraction such as $\frac{\cos(\theta + h) - \cos \theta}{h}$ or $\frac{\cos(\theta + \delta\theta) - \cos \theta}{\delta\theta}$ Allow $\frac{\cos(\theta + h) - \cos \theta}{(\theta + h) - \theta}$ o.e. Note: $\cos(\theta + h)$ or $\cos(\theta + \delta\theta)$ may be expanded
M1:	Uses the compound angle formula for $\cos(\theta + h)$ to give $\cos \theta \cos h \pm \sin \theta \sin h$
A1:	Achieves $\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h}$ or equivalent
dM1:	dependent on both the B and M marks being awarded Complete attempt to apply the given limits to the gradient of their chord
Note:	They must isolate $\frac{\sin h}{h}$ and $\left(\frac{\cos h - 1}{h}\right)$, and replace $\frac{\sin h}{h}$ with 1 and replace $\left(\frac{\cos h - 1}{h}\right)$ with 0
A1*:	cso. Uses correct mathematical language of limiting arguments to prove $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$
Note:	Acceptable responses for the final A mark include: $\frac{d}{d\theta}(\cos \theta) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta \right) = -1 \sin \theta + 0 \cos \theta = -\sin \theta$ - Gradient of chord = $-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$. As $h \rightarrow 0$, gradient of chord tends to the gradient of the curve, so derivative is $-\sin \theta$ - Gradient of chord = $-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$. As $h \rightarrow 0$, gradient of curve is $-\sin \theta$
Note:	Give final A0 for the following example which shows <i>no limiting arguments</i> : when $h = 0$, $\frac{d}{d\theta}(\cos \theta) = -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta = -1 \sin \theta + 0 \cos \theta = -\sin \theta$
Note:	Do not allow the final A1 for stating $\frac{\sin h}{h} = 1$ or $\left(\frac{\cos h - 1}{h}\right) = 0$ and attempting to apply these
Note:	In this question $\delta\theta$ may be used in place of h
Note:	Condone $f'(\theta)$ where $f(\theta) = \cos \theta$ or $\frac{dy}{d\theta}$ where $y = \cos \theta$ used in place of $\frac{d}{d\theta}(\cos \theta)$

Notes for Question Continued	
Note:	Condone x used in place of θ if this is done consistently
Note:	Give final A0 for • $\frac{d}{d\theta}(\cos x) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h} \right) \cos \theta \right) = -1 \sin \theta + 0 \cos \theta = -\sin \theta$ • $\frac{d}{d\theta} = \dots$ • Defining $f(x) = \cos \theta$ and applying $f'(x) = \dots$ • $\frac{d}{dx}(\cos \theta)$
Note:	Give final A1 for a correct limiting argument in x , followed by $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$ e.g. $\frac{d}{d\theta}(\cos x) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin x + \left(\frac{\cos h - 1}{h} \right) \cos x \right) = -1 \sin x + 0 \cos x = -\sin x$ $\Rightarrow \frac{d}{d\theta}(\cos \theta) = -\sin \theta$
Note:	Applying $h \rightarrow 0$, $\sin h \rightarrow h$, $\cos h \rightarrow 1$ to give e.g. $\lim_{h \rightarrow 0} \left(\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h} \right) = \left(\frac{\cos \theta(1) - \sin \theta(h) - \cos \theta}{h} \right) = \frac{-\sin \theta(h)}{h} = -\sin \theta$ is final M0 A0 for incorrect application of limits
Note:	$\lim_{h \rightarrow 0} \left(\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h} \right) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h} \right) \cos \theta \right)$ $= \lim_{h \rightarrow 0} (-1 \sin \theta + 0 \cos \theta) = -\sin \theta$. So for not removing $\lim_{h \rightarrow 0}$ when the limit was taken is final A0
Note:	Alternative Method: Considers $\frac{\cos(\theta+h) - \cos(\theta-h)}{(\theta+h) - (\theta-h)}$ which simplifies to $\frac{-2 \sin \theta \sin h}{2h}$

Q15.

Question	Scheme	Marks	AOs
(a)	$H = ax^2 + bx + c$ and $x=0, H=3 \Rightarrow H = ax^2 + bx + 3$	M1	3.3
	$H = ax^2 + bx + 3$ and $x=120, H=27 \Rightarrow 27 = 14400a + 120b + 3$	M1	3.1b
	or $\frac{dH}{dx} = 2ax + b = 0$ when $x = 90 \Rightarrow 180a + b = 0$	A1	1.1b
	$H = ax^2 + bx + 3$ and $x=120, H=27 \Rightarrow 27 = 14400a + 120b + 3$ and $\frac{dH}{dx} = 2ax + b = 0$ when $x = 90 \Rightarrow 180a + b = 0$ $\Rightarrow a = \dots, b = \dots$	dM1	3.1b
	$H = -\frac{1}{300}x^2 + \frac{3}{5}x + 3$ o.e.	A1	1.1b
		(5)	
(b)(i)	$x = 90 \Rightarrow H \left(= -\frac{1}{300}(90)^2 + \frac{3}{5}(90) + 3 \right) = 30 \text{ m}$	B1	3.4
(b)(ii)	$H = 0 \Rightarrow -\frac{1}{300}x^2 + \frac{3}{5}x + 3 = 0 \Rightarrow x = \dots$	M1	3.4
	$x = (-4.868\dots, 184.868\dots)$ $\Rightarrow x = 185 \text{ (m)}$	A1	3.2a
		(3)	
(c)	Examples must focus on why the model may not be appropriate or give values/situations where the model would break down: E.g. - The ground is unlikely to be horizontal - The ball is not a particle so has dimensions/size - The ball is unlikely to travel in a vertical plane (as it will spin) H is not likely to be a quadratic function in x	B1	3.5b
		(1)	
(9 marks)			
Notes			

(a)

M1: Translates the problem into a suitable model and uses $H = 3$ when $x = 0$ to establish $c = 3$

Condone with $a = \pm 1$ so $H = x^2 + bx + 3$ will score M1 but little else

M1: For a correct attempt at using one of the two other pieces of information within a quadratic model

Either uses $H = 27$ when $x = 120$ (with $c = 3$) to produce a linear equation connecting a and b for

the model Or differentiates and uses $\frac{dH}{dx} = 0$ when $x = 90$. Alternatives exist here, using the

symmetrical nature of the curve, so they could use $x = -\frac{b}{2a}$ at vertex or use point $(60, 27)$ or $(180, 3)$.

- A1: At least one correct equation connecting a and b . Remember " a " could have been set as negative so an equation such as $27 = -14400a + 120b + 3$ would be correct in these circumstances.
- dM1: Fully correct strategy that uses $H = ax^2 + bx + 3$ with the two other pieces of information in order to establish the values of both a and b for the model
- A1: Correct equation, not just the correct values of a , b and c . Award if seen in part (b)
- (b)(i)
- B1: Correct height including the units. CAO
- (b)(ii)
- M1: Uses $H = 0$ and attempts to solve for x . Usual rules for quadratics.
- A1: Discards the negative solution (may not be seen) and identifies awrt 185 m. Condone lack of units
- (c)
- B1: Candidate should either refer to an issue with one of the four aspects of how the situation has been modelled or give a situation where the model breaks down
- the ball has been modelled as a particle
 - there may be trees (or other hazards) in the way that would affect the motion
- Condone answers (where the link to the model is not completely made) such as
- the ball will spin
 - ground is not flat
- Do not accept answers which refer to the situation after it hits the ground (this isn't what was modelled)
- the ball will bounce after hitting the ground
 - it gives a negative height for some values for x
- Do not accept answers that do not refer to the model in question, or else give single word vague answers
- the height of tee may have been measured incorrectly
 - "friction", "spin", "force" etc
 - it does not take into account the weight of the ball
 - it depends on how good the golfer is
 - the shape of the ball will affect the motion
 - you cannot hit a ball the same distance each time you hit it

The method using an alternative form of the equation can be scored in a very similar way.

The first M is for the completed square form of the quadratic showing a maximum at $x = 90$

So award M1 for $H = \pm a(x-90)^2 + c$ or $H = \pm a(90-x)^2 + c$. Condone for this mark an equation with

$a = 1 \Rightarrow H = (x-90)^2 + c$ or $c = 3 \Rightarrow H = a(x-90)^2 + 3$ but will score little else

Alt (a)	$H = a(x+b)^2 + c$ and $x = 90$ at $H_{\max} \Rightarrow H = a(x-90)^2 + c$	M1	3.3
	$H = 3$ when $x = 0 \Rightarrow 3 = 8100a + c$ or $H = 27$ when $x = 120 \Rightarrow 27 = 900a + c$	M1 A1	3.1b 1.1b
	$H = 3$ when $x = 0 \Rightarrow 3 = 8100a + c$ and $H = 27$ when $x = 120 \Rightarrow 27 = 900a + c$ $\Rightarrow a = \dots, c = \dots$	dM1	3.1b
	$H = -\frac{1}{300}(x-90)^2 + 30$ o.e	A1	1.1b
		(5)	
(b)	$x = 90 \Rightarrow H = 0^2 + 30 = 30$ m	B1	3.4
		(1)	
	$H = 0 \Rightarrow 0 = -\frac{1}{300}(x-90)^2 + 30 \Rightarrow x = \dots$	M1	3.4
	$\Rightarrow x = 185$ (m)	A1	3.2a
		(2)	

Note that $H = -\frac{1}{300}(x-90)^2 + 30$ is equivalent to $H = -\frac{1}{300}(90-x)^2 + 30$

Other versions using symmetry are also correct so please look carefully at all responses

E.g. Using a starting equation of $H = a(x-60)(x-120) + b$ leads to $H = -\frac{1}{300}(x-60)(x-120) + 27$

Q16.

Question	Scheme	Marks	AOs
(a)	265 thousand	B1	3.4
		(1)	
(b)	Attempts $\frac{dN_b}{dt} = 11e^{0.05t}$	M1	1.1b
	Substitutes $t=10$ into their $\frac{dN_b}{dt}$	M1	3.4
	$\frac{dN_b}{dt} = \text{awrt } 18.1$ which is approximately 18 thousand per year *	A1*	2.1
		(3)	
(c)	Sets $45 + 220e^{0.05t} = 10 + 800e^{-0.05t} \Rightarrow 220e^{0.05t} + 35 - 800e^{-0.05t} = 0$	M1	3.1b
	Correct quadratic equation $\Rightarrow 220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$	A1	1.1b
	$e^{0.05t} = 1.829, (-1.988) \Rightarrow 0.05t = \ln(1.829)$	M1	2.1
	$T = 12.08$	A1	1.1b
		(4)	
(8 marks)			
Notes:			

(a) May be seen in the question so watch out.

B1: Accept 265 thousand or 265 000 or equivalent such as 265 k but not just 265.

(b)

M1: Differentiates to a form $ke^{0.05t}$, $k > 0, k \neq 220$. Do not be too concerned about the lhs.

M1: Substitutes $t=10$ into a changed function that was formed from an attempt at differentiation.

The left hand side must have implied differentiation. E.g. Rate = , N' , $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ or even $\frac{dy}{dx}$

A1*: Full and complete proof that requires

- some correct lhs seen at some point. E.g. "Rate = , " $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ but condone N' .
- an intermediate line/answer of either $11e^{0.05 \times 10}$ or awrt 18.1 before a minimal conclusion which must be referencing the 18 000 or 18 thousand

(c)

M1: Attempts to set both equations equal to each other and simplify the constant terms.

Look for $220e^{0.05t} + 35 = 800e^{-0.05t}$ o.e but condone slips

It is also possible to set $\frac{N-45}{220} = \left(e^{0.05t}\right) = \frac{800}{N-10}$ and form an equation in N

A1: Correct quadratic form.

Look for $220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$ or $220e^{0.1t} + 35e^{0.05t} - 800 = 0$ but allow with terms in

different order such as $220e^{0.1t} + 35e^{0.05t} = 800$

FYI the equation in N is $N^3 - 55N - 175550 = 0$

M1: Full attempt to find the value of t (or a constant multiple of t)

This involves the key step of recognising and solving a 3TQ in $e^{0.05t}$ followed by the use of lns.

If the answers to the quadratic just appear (from a calculator) you will need to check.

Accuracy should be to 3sf.

You may see different variables used such as x

$x = e^{0.05t}$, $220e^{0.1t} + 35e^{0.05t} = 800 \Rightarrow 220x^2 + 35x = 800 \Rightarrow x = 1.82... \Rightarrow t = 20 \ln 1.82...$

Allow use of calculator for solving the quadratic and for $e^{0.05t} = 1.82... \Rightarrow t = 12.08$

Via the N route it will involve substituting the positive solution to their quadratic into either equation to find a value for t/T using same rules as above.

A1: AWRT 12.08

Answers with limited or no working in (b) and (c)

(b) A derivative in the correct form must be seen

(c) Candidates who state $45 + 220e^{0.05t} = 10 + 800e^{-0.05t}$ followed by awrt 12.08 (presumably from using num-solv on their calculators) can score SC 1100. Rubric on the front of the paper states that "Answers without working may not gain full credit" so we demand a method in this part.

Q17.

Question	Scheme	Marks	AOs
(a)	$x^2 \tan y = 9 \Rightarrow 2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$	M1 A1	3.1a 1.1b
	Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$	M1	1.1b
	$\frac{dy}{dx} = \frac{-2x \times \frac{9}{x^2}}{x^2 \left(1 + \frac{81}{x^4}\right)} = \frac{-18x}{x^4 + 81} \quad *$	A1*	2.1
		(4)	
(b)	$\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{-18 \times (x^4 + 81) - (-18x)(4x^3)}{(x^4 + 81)^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2} \text{ o.e.}$	M1 A1	1.1b 1.1b
	States that when $x < \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} < 0$ when $x = \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} = 0$ AND when $x > \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} > 0$ giving a point of inflection when $x = \sqrt[4]{27}$	A1	2.4
		(3)	
		(7 marks)	
Notes:			

(a)

M1: Attempts to differentiate $\tan y$ implicitly. Eg. $\tan y \rightarrow \sec^2 y \frac{dy}{dx}$ or $\cot y \rightarrow -\operatorname{cosec}^2 y \frac{dy}{dx}$

You may well see an attempt $\tan y = \frac{9}{x^2} \Rightarrow \sec^2 y \frac{dy}{dx} = \dots$

When a candidate writes $x^2 \tan y = 9 \Rightarrow x = 3 \tan^{\frac{1}{2}} y$ the mark is scored for $\tan^{\frac{1}{2}} y \rightarrow \dots \tan^{\frac{3}{2}} y \sec^2 y$

A1: Correct differentiation $2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$

Allow also $\sec^2 y \frac{dy}{dx} = -\frac{18}{x^3}$ or $2x = -9 \operatorname{cosec}^2 y \frac{dy}{dx}$ amongst others

M1: Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$

A1*: Proceeds correctly to the given answer of $\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$

(b)

M1: Attempts to differentiate the given expression using the product or quotient rule.

For example look for a correct attempt at $\frac{vu' - uv'}{v^2}$ with $u = -18x, v = x^4 + 81, u' = \pm 18, v' = \dots x^3$

If no method is seen or implied award for $\frac{\pm 18 \times (x^4 + 81) \pm 18x(ax^3)}{(x^4 + 81)^2}$

Using the product rule award for $\pm 18(x^4 + 81)^{-1} \pm 18x(x^4 + 81)^{-2} \times cx^3$

A1: Correct simplified $\frac{d^2y}{dx^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e. such as $\frac{d^2y}{dx^2} = \frac{54x^4 - 1458}{(x^4 + 81)^2}$

Alternatively score for showing that when a correct (unsimplified) $\frac{d^2y}{dx^2} = 0 \Rightarrow x^4 = 27 \Rightarrow x = \sqrt[4]{27}$

Or for substituting $x = \sqrt[4]{27}$ into an unsimplified but correct $\frac{d^2y}{dx^2}$ and showing that it is 0

A1: Correct explanation with a minimal conclusion and correct second derivative.

See scheme.

It can be also be argued from $x^4 < 27$, $x^4 = 27$ and $x^4 > 27$ provided the conclusion states that the point of inflection is at $x = \sqrt[4]{27}$

Alternatively substitutes values of x either side of $\sqrt[4]{27}$ and at $\sqrt[4]{27}$, into $\frac{d^2y}{dx^2}$, finds all three values and makes a minimal conclusion.

A different method involves finding $\frac{d^3y}{dx^3}$ and showing that $\frac{d^3y}{dx^3} \neq 0$ and $\frac{d^2y}{dx^2} = 0$ when $x = \sqrt[4]{27}$

FYI $\frac{d^3y}{dx^3} = \frac{23328x^3}{(x^4 + 81)^3} = 0.219$ when $x = \sqrt[4]{27}$

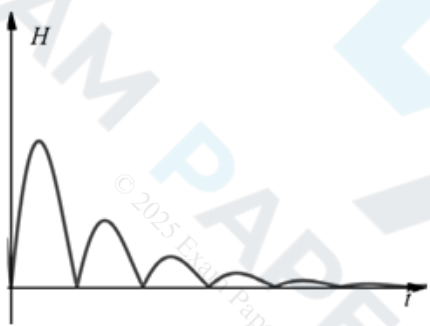
Alternative part (a) using arctan

M1: Sets $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times \dots$ where ... could be 1

A2: $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times -\frac{18}{x^3}$

A1*: $\frac{dy}{dx} = \frac{1}{1 + \frac{81}{x^4}} \times -\frac{18}{x^3} = \frac{-18x}{x^4 + 1}$ showing correct intermediate step and no errors.

Q18.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$f'(x) = -2.5e^{-0.25x} \sin x + 10e^{-0.25x} \cos x$	M1	This mark is given for a method to differentiate to find an expression for $f'(x)$
		A1	This mark is given for correctly differentiating to find an expression for $f'(x)$
	$f'(x) = 0$ $\Rightarrow e^{-0.25x} (-2.5 \sin x + 10 \cos x) = 0$ $\Rightarrow (-2.5 \sin x + 10 \cos x) = 0$	M1	This mark is given for setting $f'(x) = 0$ and finding as method to solve for $\tan x$
	$\frac{\sin x}{\cos x} = \frac{10}{2.5}$ $\tan x = 4$	A1	This mark is given for showing that $\tan x = 4$ as required.
(b)		M1	This mark is given for a graph with a correct shape
		A1	This mark is given for a graph with heights > 0
(c)	$\tan x = 4, x = 1.326$ $t = \pi + 1.326 = 4.47$	M1	This method is given for finding a value for t between the first and second bounce
	$H(4.47) = 10e^{-0.25 \times 4.47} \sin 4.47 $	M1	This mark is given for substituting the value of $t = \pi + \arctan 4$ into $H(t)$
	$= 3.27 \times -0.97 $ $= 3.17 \text{ metres}$	A1	This mark is given for finding the maximum height of the ball
(d)	The time between each bounce should not stay the same when the heights of each bounce are getting smaller	B1	This mark is given for a valid explanation of why the model should not be used to predict the time of each bounce

Q19.

Question	Scheme	Marks	AOs
	$\frac{2(x+h)^2 - 2x^2}{h} = \dots$	M1	2.1
	$\frac{2(x+h)^2 - 2x^2}{h} = \frac{4xh + 2h^2}{h}$	A1	1.1b
	$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} = \lim_{h \rightarrow 0} (4x + 2h) = 4x^*$	A1*	2.5
		(3)	
(3 marks)			
Notes:			

Throughout the question allow the use of δx for h or any other letter e.g. a if used consistently. If δx is used then you can condone e.g. $\delta^2 x$ for δx^2 as well as condoning e.g. poorly formed δ 's

M1: Begins the process by writing down the gradient of the chord and attempts to expand the correct bracket – you can condone “poor” squaring e.g. $(x+h)^2 = x^2 + h^2$.

Note that $\frac{2(x-h)^2 - 2x^2}{-h} = \dots$ is also a possible approach.

A1: Reaches a correct fraction or with the x^2 terms cancelled out.

E.g. $\frac{4xh + 2h^2}{h}$, $\frac{\cancel{2x^2} + 4xh + 2h^2 - \cancel{2x^2}}{h}$, $4x + 2h$

A1*: Completes the process by applying a limiting argument and deduces that $\frac{dy}{dx} = 4x$ with no errors seen. The “ $\frac{dy}{dx} =$ ” doesn’t have to appear but there must be something equivalent e.g. “ $f'(x) =$ ” or “Gradient =” which can appear anywhere in their working. If $f'(x)$ is used then there is no requirement to see $f(x)$ defined first. Condone e.g. $\frac{dy}{dx} \rightarrow 4x$ or $f'(x) \rightarrow 4x$.

Condone missing brackets so allow e.g. $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} = \lim_{h \rightarrow 0} 4x + 2h = 4x$

Do not allow $h = 0$ if there is never a reference to $h \rightarrow 0$

e.g. $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} = \lim_{h \rightarrow 0} 4x + 2(0) = 4x$ is acceptable

but e.g. $\frac{dy}{dx} = \frac{4xh + 2h^2}{h} = 4x + 2h = 4x + 2(0) = 4x$ is not if there is no $h \rightarrow 0$ seen.

The $h \rightarrow 0$ does not need to be present throughout the proof e.g. on every line.

They must reach $4x + 2h$ at the end and not $\frac{4xh + 2h^2}{h}$ (without the h 's cancelled) to complete the limiting argument.

Q20.

Question	Scheme	Marks	AOs
(a)	Sets $500 = \pi r^2 h$	B1	2.1
	Substitute $h = \frac{500}{\pi r^2}$ into $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \times \frac{500}{\pi r^2}$	M1	2.1
	Simplifies to reach given answer $S = 2\pi r^2 + \frac{1000}{r}$ *	A1*	1.1b
		(3)	
(b)	Differentiates S with both indices correct in $\frac{dS}{dr}$	M1	3.4
	$\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$	A1	1.1b
	Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant	M1	2.1
	Radius = 4.30 cm	A1	1.1b
	Substitutes their $r = 4.30$ into $h = \frac{500}{\pi r^2} \Rightarrow$ Height = 8.60 cm	A1	1.1b
		(5)	
(c)	States a valid reason such as - The radius is too big for the size of our hands - If $r = 4.3$ cm and $h = 8.6$ cm the can is square in profile. All drinks cans are taller than they are wide - The radius is too big for us to drink from - They have different dimensions to other drinks cans and would be difficult to stack on shelves with other drinks cans	B1	3.2a
		(1)	
9 marks			

Notes:
(a) B1: Uses the correct volume formula with $V=500$. Accept $500 = \pi r^2 h$ M1: Substitutes $h = \frac{500}{\pi r^2}$ or $rh = \frac{500}{\pi r}$ into $S = 2\pi r^2 + 2\pi r h$ to get S as a function of r A1*: $S = 2\pi r^2 + \frac{1000}{r}$ Note that this is a given answer.
(b) M1: Differentiates the given S to reach $\frac{dS}{dr} = Ar \pm Br^{-2}$ A1: $\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$ or exact equivalent M1: Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant A1: $R =$ awrt 4.30cm A1: $H =$ awrt 8.60 cm
(c) B1: Any valid reason. See scheme for alternatives

Q21.

Question	Scheme	Marks	AOs
	$\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$	M1 A1	3.1a 1.1b
	Substitutes $x = 4 \Rightarrow \frac{dy}{dx} = 6$	M1	2.1
	Uses (4, 15) and gradient $\Rightarrow y - 15 = 6(x - 4)$	M1	2.1
	Equation of l is $y = 6x - 9$	A1	1.1b
	Area $R = \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$	M1	3.1a
	$= \left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x(+c) \right]_0^4$	A1	1.1b
	Uses both limits of 4 and 0 $\left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x \right]_0^4 = 2 \times 4^{\frac{5}{2}} - \frac{15}{2} \times 4^2 + 20 \times 4 - 0$	M1	2.1
	Area of $R = 24$ *	A1*	1.1b
	Correct notation with good explanations	A1	2.5
		(10)	
(10 marks)			

Notes:

M1: Differentiates $5x^{\frac{3}{2}} - 9x + 11$ to a form $Ax^{\frac{1}{2}} + B$

A1: $\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$ but may not be simplified

M1: Substitutes $x = 4$ in their $\frac{dy}{dx}$ to find the gradient of the tangent

M1: Uses their gradient and the point (4, 15) to find the equation of the tangent

A1: Equation of l is $y = 6x - 9$

M1: Uses Area $R = \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$ following through on their $y = 6x - 9$

Look for a form $Ax^{\frac{5}{2}} + Bx^2 + Cx$

A1: $= \left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x + c \right]_0^4$ This must be correct but may not be simplified

M1: Substitutes in both limits and subtracts

A1*: Correct area for $R = 24$

A1: Uses correct notation and produces a well explained and accurate solution. Look for

- Correct notation used consistently and accurately for both differentiation and integration
- Correct explanations in producing the equation of l . See scheme.
- Correct explanation in finding the area of R . In way 2 a diagram may be used.

Alternative method for the area using area under curve and triangles. (Way 2)

M1: Area under curve $= \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) = \left[Ax^{\frac{5}{2}} + Bx^2 + Cx \right]_0^4$

A1: $= \left[2x^{\frac{5}{2}} - \frac{9}{2}x^2 + 11x \right]_0^4 = 36$

M1: This requires a full method with all triangles found using a correct method

Look for Area $R = \text{their } 36 - \frac{1}{2} \times 15 \times \left(4 - \text{their } \frac{3}{2} \right) + \frac{1}{2} \times \text{their } 9 \times \text{their } \frac{3}{2}$

Q22.

Question	Scheme	Marks	AOs
(a)	Attempts to differentiate using the quotient rule or otherwise	M1	2.1
	$f'(x) = \frac{e^{\sqrt{2}x-1} \times 8 \cos 2x - 4 \sin 2x \times \sqrt{2} e^{\sqrt{2}x-1}}{(e^{\sqrt{2}x-1})^2}$	A1	1.1b
	Sets $f'(x) = 0$ and divides/ factorises out the $e^{\sqrt{2}x-1}$ terms	M1	2.1
	Proceeds via $\frac{\sin 2x}{\cos 2x} = \frac{8}{4\sqrt{2}}$ to $\Rightarrow \tan 2x = \sqrt{2}^*$	A1*	1.1b
		(4)	
(b)	(i) Solves $\tan 4x = \sqrt{2}$ and attempts to find the 2 nd solution	M1	3.1 a
	$x = 1.02$	A1	1.1b
	(ii) Solves $\tan 2x = \sqrt{2}$ and attempts to find the 1 st solution	M1	3.1 a
	$x = 0.478$	A1	1.1b
		(4)	
(8 marks)			

Notes:

(a)

M1: Attempts to differentiate by using the quotient rule with $u = 4 \sin 2x$ and $v = e^{\sqrt{2}x-1}$ or alternatively uses the product rule with $u = 4 \sin 2x$ and $v = e^{1-\sqrt{2}x}$

A1: For achieving a correct $f'(x)$. For the product rule

$$f'(x) = e^{1-\sqrt{2}x} \times 8 \cos 2x + 4 \sin 2x \times -\sqrt{2} e^{1-\sqrt{2}x}$$

M1: This is scored for cancelling/ factorising out the exponential term. Look for an equation in just $\cos 2x$ and $\sin 2x$

A1*: Proceeds to $\tan 2x = \sqrt{2}$. This is a given answer.

(b) (i)

M1: Solves $\tan 4x = \sqrt{2}$ attempts to find the 2nd solution. Look for $x = \frac{\pi + \arctan \sqrt{2}}{4}$

Alternatively finds the 2nd solution of $\tan 2x = \sqrt{2}$ and attempts to divide by 2

A1: Allow awrt $x = 1.02$. The correct answer, with no incorrect working scores both marks

(b)(ii)

M1: Solves $\tan 2x = \sqrt{2}$ attempts to find the 1st solution. Look for $x = \frac{\arctan \sqrt{2}}{2}$

A1: Allow awrt $x = 0.478$. The correct answer, with no incorrect working scores both marks

Q23.

Question	Scheme	Marks	AOs
(a)	$(f'(x) =) 4 \cos\left(\frac{1}{2}x\right) - 3$	M1 A1	1.1b 1.1b
	Sets $f'(x) = 4 \cos\left(\frac{1}{2}x\right) - 3 = 0 \Rightarrow x =$	dM1	3.1a
	$x = 14.0$ Cao	A1	3.2a
		(4)	
(b)	Explains that $f(4) > 0$, $f(5) < 0$ and the function is continuous	B1	2.4
		(1)	
(c)	Attempts $x_1 = 5 - \frac{8 \sin 2.5 - 15 + 9}{"4 \cos 2.5 - 3"}$ (NB $f(5) = -1.212...$ and $f'(5) = -6.204...$)	M1	1.1b
	$x_1 = \text{awrt } 4.80$	A1	1.1b
		(2)	
(7 marks)			
Notes:			

(a)

M1: Differentiates to obtain $k \cos\left(\frac{1}{2}x\right) \pm a$ where a is a constant which may be zero and no other terms. The brackets are not required.

A1: Correct derivative $f'(x) = 4 \cos\left(\frac{1}{2}x\right) - 3$. Allow unsimplified e.g. $f'(x) = \frac{1}{2} \times 8 \cos\left(\frac{1}{2}x\right) - 3x^0$

There is no need for $f'(x) = \dots$ or $\frac{dy}{dx} = \dots$ just look for the expression and the brackets are not required.

dM1: For the complete strategy of proceeding to a value for x .

Look for

- $f'(x) = a \cos\left(\frac{1}{2}x\right) + b = 0$, $a, b \neq 0$
- Correct method of finding a valid solution to $a \cos\left(\frac{1}{2}x\right) + b = 0$

Allow for $a \cos\left(\frac{1}{2}x\right) + b = 0 \Rightarrow \cos\left(\frac{1}{2}x\right) = \pm k \Rightarrow x = 2 \cos^{-1}(\pm k)$ where $|k| < 1$

If this working is not shown then you may need to check their value(s).

For example $4 \cos\left(\frac{1}{2}x\right) - 3 = 0 \Rightarrow x = 1.4... \text{ or } 11.1... \text{ (or } 82.8... \text{ or } 637... \text{ or } 803 \text{ in degrees)}$ would indicate this method.

A1: Selects the correct turning point $x = 14.0$ and not just 14 or unrounded e.g. 14.011...

Must be this value only and no other values unless they are clearly rejected or 14.0 clearly selected. Ignore any attempts to find the y coordinate.

Correct answer with no working scores no marks.

(b)

B1: See scheme. Must be a full reason, (e.g. change of sign and continuous)

Accept equivalent statements for $f(4) > 0$, $f(5) < 0$ e.g. $f(4) \times f(5) < 0$, "there is a change of sign", "one negative one positive". A minimum is "change of sign and continuous" but do not allow this mark if the comment about continuity is clearly incorrect e.g. "because x is continuous" or "because the interval is continuous"

(c)

M1: Attempts $x_1 = 5 - \frac{f(5)}{f'(5)}$ to obtain a value following through on their $f'(x)$ as long as it is a

“changed” function.

Must be a correct N-R formula used – may need to check their values.

Allow if attempted in degrees. For reference in degrees $f(5) = -5.65\dots$ and $f'(5) = 0.996\dots$

and gives $x_1 = 10.67\dots$

There must be clear evidence that $5 - \frac{f(5)}{f'(5)}$ is being attempted.

so e.g. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x_1 = 4.80$ scores M0 as does e.g. $x_1 = x - \frac{8\sin\left(\frac{1}{2}x\right) - 3x + 9}{4\cos\left(\frac{1}{2}x\right) - 3} = 4.80$

BUT evidence may be provided by the accuracy of their answer. Note that the full N-R accuracy is 4.804624337 so e.g. 4.805 or 4.804 (truncated) with no evidence of incorrect work may imply the method.

A1: $x_1 =$ awrt 4.80 not awrt 4.8 but isw if awrt 4.80 is seen. Ignore any subsequent iterations.

Note that work for part (a) cannot be recovered in part (c)

Note also:

$5 - \frac{f(5)}{f'(5)} =$ awrt 4.80 following a correct derivative scores M1A1

$5 - \frac{f(5)}{f'(5)} \neq$ awrt 4.80 with no evidence that $5 - \frac{f(5)}{f'(5)}$ was attempted scores M0

Q24.

Question	Scheme	Marks	AOs
(a)	(i) $\frac{dy}{dx} = 2x - 2 - 12x^{-\frac{1}{2}}$	M1 A1	1.1b 1.1b
	(ii) $\frac{d^2y}{dx^2} = 2 + 6x^{-\frac{3}{2}}$	B1ft	1.1b
		(3)	
(b)	Substitutes $x = 4$ into their $\frac{dy}{dx} = 2 \times 4 - 2 - 12 \times 4^{-\frac{1}{2}} = \dots$	M1	1.1b
	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" oe	A1	2.1
		(2)	
(c)	Substitutes $x = 4$ into their $\frac{d^2y}{dx^2} = 2 + 6 \times 4^{-\frac{3}{2}} = (2.75)$	M1	1.1b
	$\frac{d^2y}{dx^2} = 2.75 > 0$ and states "hence minimum"	A1ft	2.2a
		(2)	
(7 marks)			

(a)(i)

M1: Differentiates to $\frac{dy}{dx} = Ax + B + Cx^{\frac{1}{2}}$ **A1:** $\frac{dy}{dx} = 2x - 2 - 12x^{\frac{1}{2}}$ (Coefficients may be unsimplified)

(a)(ii)

B1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx}$ (Their $\frac{dy}{dx}$ must have a negative or fractional index)

(b)

M1: Substitutes $x = 4$ into their $\frac{dy}{dx}$ and attempts to evaluate. There must be evidence $\frac{dy}{dx}|_{x=4} = \dots$

Alternatively substitutes $x = 4$ into an equation resulting from $\frac{dy}{dx} = 0$ Eg. $\frac{36}{x} = (x-1)^2$ and equates

A1: There must be a reason and a minimal conclusion. Allow $\checkmark$, QED for a minimal conclusion

Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" oe

Alt Shows that $x = 4$ is a root of the resulting equation and states "hence there is a stationary point"

All aspects of the proof must be correct including a conclusion

(c)

M1: Substitutes $x = 4$ into their $\frac{d^2y}{dx^2}$ and calculates its value, or implies its sign by a statement such as

when $x = 4 \Rightarrow \frac{d^2y}{dx^2} > 0$. This must be seen in (c) and not labelled (b). Alternatively calculates the gradient of C either side of $x = 4$ or calculates the value of y either side of $x = 4$.

A1ft: For a correct calculation, a valid reason and a correct conclusion. Ignore additional work where

candidate finds $\frac{d^2y}{dx^2}$ left and right of $x = 4$. Follow through on an incorrect $\frac{d^2y}{dx^2}$ but it is dependent upon having a negative or fractional index. Ignore any references to the word convex. The nature of the turning point is "minimum".

Using the gradient look for correct calculations, a valid reason.... goes from negative to positive, and a correct conclusion ...minimum.

Q25.

Question	Scheme	Marks	AOs
	$\frac{dy}{d\theta} = \frac{(2 \sin \theta + 2 \cos \theta) 3 \cos \theta - 3 \sin \theta (2 \cos \theta - 2 \sin \theta)}{(2 \sin \theta + 2 \cos \theta)^2}$	M1 A1	1.1b 1.1b
	Expands and uses $\sin^2 \theta + \cos^2 \theta = 1$ at least once in the numerator or the denominator or uses $2 \sin \theta \cos \theta = \sin 2\theta$ in $\Rightarrow \frac{dy}{d\theta} = \frac{\dots}{\dots C \sin \theta \cos \theta}$	M1	3.1a
	Expands and uses $\sin^2 \theta + \cos^2 \theta = 1$ the numerator and the denominator AND uses $2 \sin \theta \cos \theta = \sin 2\theta$ in $\Rightarrow \frac{dy}{d\theta} = \frac{P}{Q + R \sin 2\theta}$	M1	2.1
	$\Rightarrow \frac{dy}{d\theta} = \frac{3}{2 + 2 \sin 2\theta} = \frac{\frac{3}{2}}{1 + \sin 2\theta}$	A1	1.1b
(5 marks)			

Notes:

M1: For choosing either the quotient, product rule or implicit differentiation and applying it to the given function. Look for the correct form of $\frac{dy}{d\theta}$ (condone it being stated as $\frac{dy}{dx}$) but tolerate slips on the

coefficients and also condone $\frac{d(\sin \theta)}{d\theta} = \pm \cos \theta$ and $\frac{d(\cos \theta)}{d\theta} = \pm \sin \theta$

For quotient rule look for $\frac{dy}{d\theta} = \frac{(2 \sin \theta + 2 \cos \theta) \times \pm \dots \cos \theta - 3 \sin \theta (\pm \dots \cos \theta \pm \dots \sin \theta)}{(2 \sin \theta + 2 \cos \theta)^2}$

For product rule look for

$\frac{dy}{d\theta} = (2 \sin \theta + 2 \cos \theta)^{-1} \times \pm \dots \cos \theta \pm 3 \sin \theta \times (2 \sin \theta + 2 \cos \theta)^{-2} \times (\pm \dots \cos \theta \pm \dots \sin \theta)$

Implicit differentiation look for $(\dots \cos \theta \pm \dots \sin \theta) y + (2 \sin \theta + 2 \cos \theta) \frac{dy}{d\theta} = \dots \cos \theta$

A1: A correct expression involving $\frac{dy}{d\theta}$ condoning it appearing as $\frac{dy}{dx}$

M1: Expands and uses $\sin^2 \theta + \cos^2 \theta = 1$ at least once in the numerator or the denominator OR uses

$2 \sin \theta \cos \theta = \sin 2\theta$ in $\Rightarrow \frac{dy}{d\theta} = \frac{\dots}{\dots C \sin \theta \cos \theta}$

M1: Expands and uses $\sin^2 \theta + \cos^2 \theta = 1$ in the numerator and the denominator AND uses

$2 \sin \theta \cos \theta = \sin 2\theta$ in the denominator to reach an expression of the form $\frac{dy}{d\theta} = \frac{P}{Q + R \sin 2\theta}$

A1: Fully correct proof with $A = \frac{3}{2}$ stated but allow for example $\frac{\frac{3}{2}}{1 + \sin 2\theta}$

Allow recovery from missing brackets. Condone notation slips. This is not a given answer